Prabhavit Gupta



Forward Deployed Engineer: Applied AI, Agent Architectures & Production Client Integrations

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Summary

Forward Deployed AI Engineer with 3 years of hands-on software and systems experience designing, deploying, and embedding production-grade AI solutions, RAG pipelines, and full-stack integrations. Proven track record partnering directly with enterprise and client stakeholders to unpack complex business workflows, architect scoped technical solutions in days, and deliver hardened, observable cloud deployments in weeks. Deep technical foundation in Python, TypeScript/JavaScript, PostgreSQL, and GCP/cloud architectures, with practical specialization in applied generative AI: multi-step agentic orchestration, Model Context Protocol (MCP), tool/function calling, semantic retrieval, and real-world transcript evaluation. Native AI-first practitioner operating daily with Claude Code, Gemini CLI, and automated testing (Playwright) to build systems that remain maintainable after client handover. Holder of EU Long-Term Residence Permit (permanent unrestricted EU work authorization without visa sponsorship); based in Venice, Italy and ready for immediate relocation to Utrecht / the Netherlands.

Technical Expertise

- **Forward Deployed & Solutions Delivery:** Rapid client discovery, technical scoping, stakeholder management, translating enterprise pain points into technical architecture, integration runbooks, production handover.
- **Applied AI & Agent Architectures:** Multi-step agent orchestration, tool/function calling, Model Context Protocol (MCP), agent-to-agent interoperability, schema-enforced JSON validation, guardrails, fallback escalation.
- **RAG & Retrieval Systems:** Chunking strategies, vector embeddings, hybrid semantic/keyword search, Vertex AI Search / RAG pipelines, BigQuery integrations, knowledge grounding against dynamic databases.
- **Full-Stack & Backend Engineering:** Python (FastAPI, backend services), TypeScript / JavaScript (Node.js, Next.js, React), RESTful APIs, webhook synchronization, automated contract and regression testing (Playwright).
- **Data Pipelines & Cloud Infrastructure:** PostgreSQL, schema design, Supabase, Google Cloud Platform (GCP), Docker containers, serverless compute, workflow automation (n8n), logging and production observability.
- **AI-Native Development & Evals:** Daily power user of Claude Code, Gemini CLI, and Cursor; production evaluation loops, latency/cost profiling, and regression suites over real operational logs.
- **Work Authorization & Languages:** English (C1 Professional), Hindi (Native), Italian (Beginner). EU Long-Term Residence Permit (unrestricted work authorization across the EU, no visa sponsorship required).

Experience

FlowSpace.co **Mar 2024 - Aug 2026**
Lead Developer (Full-time, Freelance) *London, UK (remote)*

- Architected, deployed, and maintained the production AI customer agent on a live platform serving 4,500+ active users, automating customer inquiries across pricing, venue inventory, and bookings.
- Grounded conversational agents against operational databases and partner APIs (OfficeRnD, Nexudus) using schema-enforced tool-calling, establishing fallback escalation triggers when model confidence dipped below thresholds.
- Established automated evaluation pipelines analyzing real production transcripts to detect failure modes and hallucination risks, driving answer precision and eliminating incorrect pricing outputs.
- Profiled latency and token economics across model iterations, optimizing prompt templates and context window utilization to reduce inference costs while keeping p95 response latencies low.
- Designed and shipped automated regression and integration test suites using Playwright, verifying API contracts and continuous deployment stability across end-to-end customer booking flows.
- Spearheaded platform migration from legacy architecture to a modern Next.js and backend service stack, supporting growth from 2,000 to 4,500+ users without operational disruption.
- Implemented data ingestion pipelines parsing, normalizing, and verifying unstructured partner venue listings, expanding platform inventory to over 900 locations.

Urlink **Apr 2025 - Jul 2025**
Project Manager & AI Assisted Developer (Freelance) *Barcelona, Spain (remote)*

- Architected core web application backend workflows, defining clear API contracts, relational schemas, and data transformation logic for client-facing features.

- Implemented automated validation and synchronization pipelines connecting third-party services with PostgreSQL, preventing malformed inputs and ensuring relational data integrity.
- Integrated automated CI/CD pipelines and regression suites, accelerating release cadences while maintaining high system stability.

Independent Technical Consultant **Aug 2023 - Feb 2024**
AI Solutions & Automation Engineer *Venice, Italy*

- Designed and delivered bespoke AI automation workflows, web extraction engines, and API integrations for European SME clients using Python, TypeScript, and Playwright.
- Translated manual, fragmented business operations into automated software solutions with structured error logging, retry queues, and reliable recovery logic.
- Engineered webhook listeners and real-time synchronization pipelines across CRM platforms, operational tools, and relational datastores.

Education

Ca' Foscari University of Venice **2020 - 2024**
BA in Philosophy, International Studies and Economics *Venice, Italy*

- Rigorous grounding in formal logic, analytical problem solving, systems analysis, and macroeconomic decision models.